INCH-POUND
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DETAIL SPECIFICATION

VALVES, QUARTER TURN, BUTTERFLY, HIGH PERFORMANCE (DOUBLE-OFFSET) WAFER AND LUG STYLE, CLASS 150, FOR SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers double-offset (see 6.3.3) position seated, class 150, quarter turn high performance wafer and lug style butterfly valves for fuel oil, aviation fuel, lubricating oil, seawater, auxiliary seawater, freshwater, chilled water, and potable water piping system services for shipboard use. Associated manual operators (gearbox) and electric, hydraulic, or pneumatic actuators are covered by this specification in limited terms.

1.2 Classification. High performance butterfly valves are of the following styles, types, and sizes identified below (see 6.2).

1.2.1 Body styles. The following high performance butterfly valve body styles are covered by this specification:

a. Style A – Wafer (see 6.3.6)

b. Style B – Lug (see 6.3.5)

1.2.2 Types. High performance butterfly valves are classified into the following four types based on material of manufacture and seating type:

a. Type I – Corrosion-resistant steel, synthetic seated, fire resistant

b. Type II – Corrosion-resistant steel, metal seated

c. Type III – Non-ferrous, synthetic seated, fire resistant

d. Type IV – Non-ferrous, metal seated

1.2.3 Sizes. High performance butterfly valve sizes conform to the following nominal pipe sizes (NPS) in inches:

a. 2	f. 5	k. 14
b. 2½	g. 6	l. 16
c. 3	h. 8	m. 18
d. 3½	i. 10	n. 20
e. 4	j. 12	o. 24

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plates, Identification
MIL-DTL-16884	-	Fuel, Naval Distillate
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service
MIL-C-24723	-	Castings, Nickel-Copper Alloy

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-798	-	Design Evaluation Requirements for Valves for Naval Shipboard Use

(Copies of these documents are available online at https://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385620 - Handwheels for Valves

803-1385892 - Flanges, Bronze, Special, for Use With Butterfly Valves

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of this document are available online via TDMIS at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24 Metric/Inch Standard

ASME B16.34 - Valves Flanged, Threaded, and Welding End

ASME B18.2.1 - Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

(Copies of these documents are available online at www.asme.org/.)

AMERICAN PETROLEUM INSTITUTE (API)

API-STD-598 - Valve Inspection and Testing

API-STD-607 - Fire Test for Quarter-Turn Valves and Valves Equipped with Nonmetallic Seats

(Copies of these documents are available online at www.api.org.)

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ASTM INTERNATIONAL

ASTM A276/A276M	-	Standard Specification for Stainless Steel Bars and Shapes
ASTM A479/A479M	-	Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels
ASTM A494/A494M	-	Standard Specification for Castings, Nickel and Nickel Alloy
ASTM A564/A564M	-	Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B62	-	Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B150/B150M	-	Standard Specification for Aluminum Bronze Rod, Bar, and Shapes
ASTM B164	-	Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire

(Copies of these documents are available online at www.astm.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ISA 75.02.01	-	Control Valve Capacity Test Procedures
NSF/ANSI/CAN 61	-	Drinking Water System Components – Health Effects

(Copies of these documents are available online at http://webstore.ansi.org/.)

FLUID CONTROLS INSTITUTE (FCI)

FCI 70-2	-	Control Valve Seat Leakage

(Copies of this document are available online at www.fluidcontrolsinstitute.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Valves furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

3.2 Materials. Valve assembly material shall conform to types I and II, or types III and IV in accordance with table I.

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TABLE I. Valve assembly material.

Name of part	Type I and II material	Type III and IV material
Body	316 corrosion-resistant steel	ASTM B150/B150M, UNS C63200 MIL-B-24480, UNS C95800
Disc	316 corrosion-resistant steel	QQ-N-281, class A ASTM A494/A494M, M30C or MIL-C-24723, composition M30C ASTM B164
Stem	ASTM A276/A276M, type XM-19 ASTM A479/A479M, type XM-19 ASTM A564/A564M, type 630 1/	QQ-N-286
Stem to disc fasteners	ASTM A276/A276M, type XM-19 ASTM A479/A479M, type XM-19 ASTM A564/A564M, type 630 1/	QQ-N-286
Other wetted and moving parts	Corrosion-resistant steel chosen for strength and galling resistance	MIL-B-24480, UNS C95800 ASTM B61 QQ-N-281, class A QQ-N-286 ASTM A494/A494M, M30C or MIL-C-24723, composition M30C ASTM B164 ASTM B150/B150M, UNS C63200
Seat and stem seals	PTFE 2/, glass filled PTFE, fluorocarbon, graphitic packing, corrosion-resistant steel	PTFE, glass filled PTFE, fluorocarbon, graphitic packing, nickel-copper (Ni-Cu)
Actuator parts exposed to atmosphere	Corrosion-resistant steel, steel, ductile iron, nodular iron ASTM B61 ASTM B62 MIL-B-24480, UNS C95800 ASTM B150/B150M, UNS C63200	ASTM B61 ASTM B62 MIL-B-24480, UNS C95800 ASTM B150/B150M, UNS C63200
Manual actuator shafts	Corrosion-resistant steel Nickel-copper	Nickel-copper
Brackets, adapters, and actuator-to-valve bracket fasteners	316 corrosion-resistant steel ASTM B150/B150M, UNS C63200 MIL-B-24480, UNS C95800 Nickel-copper	ASTM B150/B150M, UNS C63200 MIL-B-24480, UNS C95800 Nickel-copper 316 corrosion-resistant steel

NOTES:

1/ Minimum tensile strength - 125 kilopounds per square inch (KSI); minimum yield strength - 105 KSI; minimum elongation - 12%.

2/ PTFE is Polytetrafluoroethylene.

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3.2.1 Material assessment. All materials that may come in contact with potable water for human consumption shall be in accordance with NSF/ANSI/CAN 61.

3.2.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. High performance butterfly valves shall be in accordance with ASME B16.34, class 150. Valves shall be of the body styles, types, seat styles, sizes, and flange styles specified (see 1.2 and 6.2). The stem, disc, and seat shall have a double-offset (see 6.3.3) design.

3.3.1 Valve body. Valve face-to-face dimensions shall be in accordance with table II. Valve size shall be as specified (see 6.2). The dimensions of types I through IV valve bodies shall be in accordance with ASME B16.34 for class 150 valves. The minimum wall thicknesses for types I through IV valves shall be calculated in accordance with the "Basis Equations for Minimum Wall Thickness" table of ASME B16.34 for class 150 valves. Note that the design criteria specified in ASME B16.34 applies only to ferrous materials and shall be modified to consider the stress limits for non-ferrous materials prior to use in the design of types III and IV (non-ferrous) valves. Valves shall be designed for installation between flanges in accordance with ASME B16.5, class 150 for ferrous (types I and II) valves and Drawing 803-1385892 for non-ferrous (types III and IV) valves.

TABLE II. Valve face-to-face dimensions.

Valve size (inches)	**Valve face-to-face (inches ±0.0625 inch)**
2	1.75
2½	1.875
3	1.875
3½	2
4	2.125
5	2.25
6	2.25
8	2.5
10	2.8125
12	3.1875
14	3.625
16	4
18	4.5
20	5
24	6.0625

3.3.2 Actuator mounting bracket. The valve body shall have integral provisions for mounting the manual operator or motor actuator. An adaptive mounting bracket may be utilized to attach the actuator, provided it is approved by NAVSEA and the entire assembly, valve, mounting bracket, and actuator are successfully shock tested and vibration tested as specified (see 3.5.3 and 3.5.4). Requirements for mounting actuators, including bolting size and pattern, shall be as specified (see 6.2).

3.3.3 Lug-style valves. Lug-style valves shall be provided with tapped holes to permit attachment to mating flanges with fasteners in accordance with 3.3.3.1.

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3.3.3.1 Fasteners. Flange fasteners for lug-style valves shall be cap screws in accordance with ASME B18.2.1. Lug-style valves shall have tapped flange holes that provide for thread engagement of fully formed threads to a depth not less than the nominal bolt diameter.

3.3.3.2 Fastener prohibitions. Tapered thread plugs or fasteners shall not be used.

3.3.4 Travel stops. Valves shall include an internal travel stop in the closed position to prevent disc overtravel. The internal stop shall be integrally cast, forged, or welded and shall be a part of the body of the valve. The stop shall be sufficiently rugged to withstand erosion by fluid flow.

3.3.5 Stem, disc, and seat assembly design. Stem, disc, and seat assembly shall be a double-offset design and shall have a single stuffing box stem seal.

3.3.6 Seats. Seats shall be designed to seal bi-directionally. Synthetic seats shall include a metal-to-metal backup seat for fire resistance.

3.3.6.1 Seating range. In the closed position of the valve, the seat shall seal over a minimum 3-degree included angle of rotation of the disc for valve size 12 inches and less, and over a 2-degree angle of disc rotation for valve size larger than 12 inches.

3.3.6.2 Seat maintenance. Seats shall be replaceable without removing the stem and disc assembly.

3.3.6.3 Seat design. Seats shall be either reversible or shall be designed to physically preclude improper installation.

3.3.6.4 Seat retainer. The seat retainer shall be contained by a recess in the body of the valve to preclude a potential external leak path through the body, and the recess shall be within the flange gasket area. The seat retainer shall not protrude beyond the body face nor be so deeply recessed that a spiral wound or other gasket is ineffective over the sealing surface. The seat retainer shall be attached to the body of the valve through the face of the retainer. Style B lug valves shall be designed such that the seat retainer can withstand the rated pressure of the valve in open end connections.

3.3.6.5 Seat material compatibility. Synthetic seating materials shall be compatible with fuels in accordance with MIL-DTL-5624 and MIL-DTL-16884 (see table I).

3.3.7 Stems. Stems shall be rigidly attached to the disc. Means shall be provided to preclude stem or stem piece expulsion from the body of the valve in the event of part failure. The stem retention device shall be in the body neck or shall be so constructed that it is functional during stem seal replacement and adjustment and cannot be disassembled during stem seal maintenance. This feature shall be independent of the valve actuating mechanism.

3.3.7.1 Stem packing. Stem packing shall be replaceable without removing the valve from the piping system.

3.3.7.2 Stem packing fire resistance. Stem packing shall incorporate a fire resistant feature.

3.3.8 Position indication. Valves shall be provided with a local position indicator that shall be an integral part of the stem section that is visible. The indicator may include a machined mark on the stem, pointer, or other means that is visible when the valve is attached to the actuator.

3.3.9 Actuators. Valves shall be actuated by manual, electric, pneumatic, or hydraulic actuators. Manual actuators for all size valves shall be enclosed worm gear type. Manual actuator handwheels shall be in accordance with Drawing 803-1385620 when the depicted sizes are used. Actuator materials shall be in accordance with table I. The type of actuator shall be as specified (see 6.2). The maximum allowable tangential force required to close the valve, based on handwheel size, shall be in accordance with table III.

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TABLE III. Maximum tangential handwheel force.

Handwheel diameter (inches)	Total tangential force on rim of handwheel (pounds)
Below 5	30
5	50
6	72
7	90
8	102
9	114
10	120
11	129
12	135
14	138
16	141
18	144
21	147
24 and above	150

3.3.9.1 Actuation time. Regardless of actuation, valve time to open or to close shall not exceed 35 seconds.

3.3.10 Special tools. Valves shall be constructed so that special tools shall not be required for installation and maintenance. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

3.3.11 Welding. Welding shall be in accordance with S9074-AR-GIB-010/278. Welds shall be free of cracks.

3.4 Physical requirements.

3.4.1 Pressure-temperature ratings. The pressure-temperature rating for types I and II high performance butterfly valves shall be in accordance with the "Ratings for Group 2.2 Materials" table in ASME B16.34 for class 150, standard class valves. Types III and IV shall be calculated in accordance with the ASME B16.34 non-mandatory Appendix B method used for establishing pressure–temperature ratings, method for class 150 – All Materials. The pressure-temperature rating for all valve types shall be rated at 100 °F.

3.4.2 Seat leakage. Types I and III valves shall have no visually detectable leakage as allowed for resilient seat leakage criteria in accordance with API-STD-598. Types II and IV metal-to-metal seats shall have a maximum seat leakage rate of 10 cubic centimeters (cm^3) per inch of nominal size per hour (see 4.5.2.2).

3.4.3 Valve body marking. Valves shall have the ASME class, the name or trademark of the manufacturer, and the recommended flow direction (see 6.3.4) arrow cast or forged integrally with the body, or permanently attached to the body.

3.4.4 Identification plate. Valves shall be provided with a permanently attached identification plate in accordance with MIL-DTL-15024 and MIL-P-15024/5, type A, B, C, D, or L. The identification plate shall contain the following information:

a. Manufacturer's name or trademark.

b. Valve size and rating per ASME B16.34.

c. Body, trim, and seat material.

d. Manufacturer's identification number (Commercial and Government Entity [CAGE] code).

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e. "MIL-DTL-24624" (military specification number).

f. Component Identification Number (CID).

3.5 Performance requirements.

3.5.1 Flow capacity. The minimum flow capacity (C_v [see 6.3.1]) shall be in accordance with table IV and shall be established in accordance with the tests of 4.5.5.

TABLE IV. Minimum flow capacity.

Valve size (inches)	C_v
2	55
2½	75
3	165
3½	250
4	400
5	600
6	950
8	1,800
10	2,900
12	4,200
14	5,800
16	7,300
18	9,600
20	13,500
24	21,600

3.5.2 Valve operation. Valves shall be capable of fully opening from the fully closed position in 35 seconds or less. Valves shall also be capable of fully closing from the fully opened position in 35 seconds or less.

3.5.3 Shock resistance. Valves, with actuators installed (gear box, electric, pneumatic, or hydraulic), comprise the principal unit and shall withstand the grade A, class I, type A shock test requirements of MIL-DTL-901 when tested as specified in 4.5.3.

3.5.4 Vibration resistance. Valves, with actuators installed (gear box, electric, pneumatic, or hydraulic), shall meet the vibration requirements of MIL-STD-167-1 when tested as specified in 4.5.4.

3.5.5 Life cycle and throttling. The valves shall be designed for a life of 20,000 cycles (see 6.3.2) and shall be capable of throttling in the 15 to 17 percent open position for a minimum of 12 hours. The valves shall pass the tests of 4.5.6.

3.5.6 Fire resistance. Synthetic seated valves shall be fire resistant and shall meet the allowable through valve and external leakage rate acceptance criteria when tested in accordance with API-STD-607 (see 4.5.7).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 Qualification inspection. Qualification inspection shall include the examination specified in 4.4 and tests specified in 4.5.2 through 4.5.8 (see table V) conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall be conducted with style A, type I and type III valves. Unless otherwise specified (see 6.2), qualification tests shall be conducted with a gearbox operator. Valve test sizes shall be 4, 8, 14, and 24 inches or as specified (see 6.2). Valve test sizes may vary for shock testing (see 4.5.3), vibration testing (see 4.5.4), and fire testing (see 4.5.7).

4.2.1 Extension of qualification. Successful completion of the tests as defined in 4.2 will allow qualification approval to be extended to the smaller intermediate valve sizes and to style B lug valves, provided that the geometry of the valve and the actuator assemblies are scaled versions of the tested valves. Type I valves may be extended to type II, and type III may be extended to type IV, as approved by NAVSEA. All extension requests are subject to NAVSEA/Designated Approval Authority (DAA) approval.

4.2.2 Use of valves with automated actuators. When electric, pneumatic, or hydraulic actuators are specified (see 3.3.9), the shock test (see 4.5.3) and vibration test (see 4.5.4) shall be performed for the smallest valve actuator assembly and largest valve actuator assembly when an identical actuator is used. Successful test results may be extended to combinations of the same actuator mounted on intermediate size wafer style valves and to the lug valve versions of the tested and extended assemblies. All extension requests are subject to NAVSEA/DAA approval.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5.2 and 4.5.8 (see table V).

TABLE V. Qualification and conformance inspections.

Examination/Test	Requirement	Verification	Qualification	Conformance
Material and design	3.2 through 3.3.11 3.4.3 and 3.4.4	4.4	X	X
Hydrostatic shell	3.4.1	4.5.2.1	X	X
Hydrostatic seat	3.4.2	4.5.2.2	X	X
Shock	3.5.3	4.5.3	X	
Vibration	3.5.4	4.5.4	X	
Flow capacity	3.5.1	4.5.5	X	
Life cycle and throttling	3.5.2 and 3.5.5	4.5.6	X	
Fire	3.5.6	4.5.7	X	
Functional check	3.5.2	4.5.8	X	X

4.4 Examination. Valves shall be examined for compliance with the requirements specified in table V. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. Any redesign or modification to the manufacturer's qualified product that is not approved by NAVSEA or re-qualified in accordance with 4.2 shall constitute rejection of the valves.

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4.5 Tests.

4.5.1 Test valve sizes. Valves of each of the 4-inch, 8-inch, 14-inch, and 24-inch sizes shall be subjected to the tests specified in 4.5.2 through 4.5.8. The 4-inch, 8-inch, 14-inch, and 24-inch sizes shall be used in lieu of the valve sizes specified in MIL-STD-798. Successful testing of the 4-inch size approves the 2-inch, 2½-inch, 3-inch, 3½-inch, and 4-inch sizes only. Successful testing of the 8-inch size approves the 5-inch, 6-inch, and 8-inch sizes only. Successful testing of the 14-inch size approves the 10-inch, 12-inch, and 14-inch sizes only. Successful testing of the 24-inch size approves the 16-inch, 18-inch, 20-inch, and 24-inch sizes only. Prior NAVSEA approval is required for qualification of any other sizes. For shock and vibration test valve sizes, see 4.5.3 and 4.5.4. For fire test valve sizes, see 4.5.7.

4.5.2 Pressure tests.

4.5.2.1 Hydrostatic shell test. Each valve shall be subjected to a body strength and mechanical assembly test at 150 percent of the ambient temperature pressure rating. The test duration shall be a minimum of 3 minutes. The valve shall be in the open position. No external leakage or weepage is allowed. Test media shall be clean fresh ambient temperature water.

4.5.2.2 Hydrostatic seat test. Following the hydrostatic shell test, the valve seat shall be subjected to 110 percent of the ambient temperature pressure rating in each direction. The non-pressurized side shall be open to atmosphere. The leakage rate shall be in accordance with 3.4.2. The test duration shall be a minimum of 1 minute. Test media shall be clean fresh ambient temperature water.

4.5.3 Shock. Valves, with actuators installed (gear box, electric, pneumatic, or hydraulic), shall be subjected to a shock test to determine conformance with the requirements specified in 3.5.3. Test assemblies shall be tested in accordance with the grade A, class I, type A shock test requirements of MIL-DTL-901. Valve sizes used for shock testing may differ from those specified in 4.5.1. Shock test sizes shall use the smallest valve and largest valve used with an actuator/operator of the identical type/model. When more than one type/model of actuator/operator is required to drive all sizes, each actuator/operator type/model shall be shock tested using the smallest and largest valve for that particular actuator/operator type/model. For purposes of the shock test, the following MIL-DTL-901 classifications shall be applied to the valve assembly:

a. Mounting location – hull and deck mounted as specified (see 6.2).

b. Mounting plane – other, in-line pipe.

c. Mounting orientation – unrestricted.

d. Test category – dependent on shock machine weight limitations for hull and deck mounted valves.

During the shock test, the valve shall be pressurized at the 100 °F rated pressure with ambient temperature water. The valve shall be tested both open with pressure throughout and closed with pressure on the preferred side with downstream side open to atmosphere. Unless otherwise specified (see 6.2), valve closed is considered the primary mode of operation; the secondary mode is valve open.

4.5.3.1 Initial post-shock test. The initial post-shock test shall consist of a functional check in accordance with 4.5.8 and an external visual and dimensional check without disassembling the valve and hydrostatic pressure tests in accordance with 4.5.2. External visual inspection should look for items such as cracking, deformation, misalignment, and loose parts.

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4.5.3.2 Final post-shock inspection. Tested valves shall be disassembled and inspected for breakage, cracking, deformation, and misalignment. Highly stressed areas and seating areas to include the transition areas, both internally and externally, between the valve body and the stem housing, the stem housing and the operator bracket, the entire length of the stem, and the seating areas shall be checked for cracks using a liquid penetrant test in accordance with S9074-AR-GIB-010/278. A dimensional check of the valve and valve internals shall be performed to determine valve damage. The valve and valve actuator shall meet the requirements of 3.5.3. When considering non-favorable results for liquid penetrant testing, higher consideration shall be given to successful passage of post-shock seat tightness, hydrostatic shell testing, and the functional check.

4.5.4 Vibration. Valves shall be subjected to a vibration test to determine conformance with the requirements specified in 3.5.4. Valves, with actuators installed, shall be tested in accordance with the type I requirements of MIL-STD-167-1. Testing shall be conducted while the valve is pressurized at 100 °F rated pressure with ambient temperature water, once in the closed position, then in the open position. There shall be no evidence of valve structural damage or performance degradation. Valve sizes used for vibration testing may differ from those specified in 4.5.1. Vibration test sizes shall use the smallest valve and largest valve used with an actuator/operator of the identical type/model. When more than one type/model of actuator/operator is required to drive all sizes, each actuator/operator type/model shall be vibration tested using the smallest and largest valve for that particular actuator/operator type/model.

4.5.4.1 Post-vibration test. The initial post-vibration test shall consist of a functional check in accordance with 4.5.8 and an external visual and dimensional check without disassembling the valve and hydrostatic pressure tests in accordance with 4.5.2. External visual inspection should look for items such as cracking, deformation, misalignment, and loose parts.

4.5.5 Flow capacity test. Flow tests shall be conducted in accordance with ANSI/ISA 75.02.01 to determine conformance with 3.5.1. Flow capacities of the intermediate smaller sizes may be extrapolated provided the flow path geometry is a scaled version of the test valve. Flow direction during flow capacity testing shall be from the most disadvantageous side (smaller opening) to larger opening.

4.5.6 Life cycle and throttling tests. The following life cycle and throttling tests shall be conducted to determine conformance with the requirements of 3.5.5. The sequence of these tests shall be determined by the manufacturer.

4.5.6.1 Life cycle test. The life cycle test shall be conducted using a test fixture that shall allow either side of the valve to be pressurized at the 100 °F rated pressure with clean fresh water when the valve is in the closed position. Once a side is pressurized, the valve shall be cycled to simulate operation in a shipboard system. The valve shall undergo 20,000 cycles. Every 2,000 cycles, the valve shall be tested for leakage in both the preferred side and non-preferred side directions using clean fresh water at the 100 °F rated pressure. The test pressure shall be held for a minimum of 1 minute. The number of cycles and the associated leakage rate shall be recorded. Every 2,000 cycles, the flow direction shall be reversed to verify the bi-directional capabilities of the valve. The operating time shall not exceed 35 seconds. The seat and stem seal shall not be replaced during the test. During the cycle test, the stem seal may be adjusted. Upon completion of the cycle tests, a seat leakage test shall be conducted in accordance with and shall conform to the acceptance criteria of FCI 70-2, class 4. Seizing of the moving parts shall constitute failure of the test.

4.5.6.2 Throttling test. Each test valve shall be set in the 15 to 17 percent open position and clean fresh water shall be passed through the valve for a minimum of 12 hours. 4-, 8-, and 14- inch size valves shall be throttled through a 100-pound per square inch (lb/in^2) differential, and 24-inch size valves shall be throttled through a 50-lb/in^2 differential. A new seat may be used for this test. Upon completion of the throttling test, a seat leakage test shall be conducted in accordance with and shall conform to the acceptance criteria of FCI 70-2, class 4.

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4.5.7 Fire tests.

4.5.7.1 Procedure. Type I and type III valves shall be fire tested in accordance with API-STD-607. Test sizes shall be selected from the "Other Valves Qualified by NPS" table of API-STD-607. When selecting test sizes, consideration should be given to selecting the least number of valve sizes for fire testing to encompass a range of sizes of the same basic design that require qualification. During the test, type I valves shall be mated to flanges in accordance with ASME B16.5, class 150 raised face flanges. Type III valves shall be mated to flanges in accordance with Drawing 803-1385892. Leakage shall be measured by monitoring the source tank water levels during the test.

4.5.7.2 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by NAVSEA. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.5).

4.5.8 Functional check. Each valve, with the actuator installed, shall be cycled a minimum of three times to demonstrate proper functioning of the assembly. Cycling shall be accomplished using each means of operation capable by the principal unit. Cycling times shall meet the requirements of 3.3.9.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valves are military grade capable of withstanding a shock event and continuing to function after the event. The valves are intended for use in water, fuel, and oil services in naval shipboard piping systems. Existing stock of butterfly valves manufactured to MIL-V-24624 of 3 November 1983 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Style, type, and size required (see 1.2 and 4.2).
c. Requirements for mounting actuators (see 3.3.2).
d. Type of actuator required (see 3.3.9 and 4.2).
e. Mounting location (see 4.5.3.a).
f. Valve mode of operation (see 4.5.3).
g. Additional fire testing provisions (see 4.5.7.2 and 6.5).
h. Packaging requirements (see 5.1).
i. Provisioning requirements (see 6.6).

6.3 Definitions.

6.3.1 C_v. The C_v is the number of United States gallons of water per minute that the valve will pass with a differential pressure of 1 pound per square inch.

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6.3.2 Cycle. A cycle consists of fully opening the valve from the fully closed position and returning it to the fully closed position.

6.3.3 Double-offset. Characterized by offset of the valve stem behind the centerline of the valve sealing surface and an offset of the stem to the side of the centerline of the pipe. The combination of these two offsets creates a camming effect as the disc swings into and out of the seat.

6.3.4 Flow direction. Although the valve is designed for bi-directional flow and seat tightness, the preferred direction of flow is recommended by the valve manufacturer and indicated on the valve body marking (see 3.4.3).

6.3.5 Lug-style butterfly valve. Lug-style valves have threaded inserts at both sides of the valve body. This allows the valve to be installed into a system using two sets of bolts and no nuts. The valve is installed between two flanges using a separate set of bolts for each flange. This setup permits either side of the piping system to be disconnected without disturbing the other side.

6.3.6 Wafer-style butterfly valve. The wafer-style butterfly valve is designed to be installed between two flanges using a cross bolt pattern. There are no predrilled inserts. Bolts go around the outer diameter of the valve body. Bolting can be one continuous threaded stud with two nuts or one bolt and one nut for each hole on the mated flanges.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24624 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4.1 Provisions governing qualification. Copies of SD-6, “Provisions Governing Qualification” are available online at https://quicksearch.dla.mil.

6.5 Additional fire testing provisions. NAVSEA reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. A2LA (www.A2LA.org) or International Code Council’s IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified (see 6.2). When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and verifications as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any valve under this specification:

a. Engineering drawings.
b. Materials listing.
c. Installation drawings.
d. Technical manual.
e. Test reports.
f. Certifications.
g. Onboard repair parts list.

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6.8 Subject term (key word) listing.

Actuator

Bronze

Corrosion-resistant steel

Fire resistant

Synthetic seated

Throttling

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.10 Disposition of tested items. Tested items are allowed for shipboard use after refurbishment, but not allowed as sea valves or hull penetration valves after shock testing.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – CC

Preparing activity:
Navy – SH
(Project 4820-2019-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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